

Customer Churn Analysis — Detailed Project Report

Executive Summary

This project analyzes customer attrition using demographic, subscription, and customer support datasets stored in an SQLite database. The primary goal is to identify churn drivers, calculate core business Key Performance Indicators (KPIs), perform feature engineering, and create correlation heatmaps and visualizations to guide customer retention strategies.

Project Objectives

- **Data Integration & Cleaning:** Extract relational tables from an SQLite database (`customer_churn.db`), clean missing values, standardise categorical variables, and align data types.
 - **Feature Engineering:** Create derived attributes such as `churn_flag` , `complaint_count` , `tenure_days` , and categorical `churn_risk` segments.
 - **Exploratory Data Analysis (EDA):** Calculate critical financial and operational KPIs (e.g., Churn Rate, Retention Rate, ARPU, Revenue at Risk, Escalation Rate).
 - **Visualization & Pattern Recognition:** Leverage Seaborn and Matplotlib to analyze churn trends across plan types, geographic regions, and support interactions.
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Technology Stack

- **Database:** SQLite3 (`customer_churn.db`)
 - **Programming Languages:** Python 3.10
 - **Data Analysis & Processing:** Pandas, NumPy
 - **Data Visualization:** Matplotlib, Seaborn
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System Architecture & Data Pipeline Workflow

SQLite Database

db_customer

db_subscription

db_support



Data Cleaning

- Column renaming & missing value imputation
- Gender standardization ('Men'/'Women' → 'Male'/'Female')
- Date parsing (`dob`, dates to `datetime64`)
- Unused column removal (`interests`, `pincode`, etc.)



Feature Engineering

- `churn\_flag` generation from `cancellation\_date`
- Deduplication & calculation of `complaint\_count`
- Calculation of `tenure\_days`
- Discretization of `churn\_score` into `churn\_risk`



Data Merging & Export

- Left-join on `customerid` to create master dataframe
- Export to `exported\_churn\_data.csv`



Exploratory Data Analysis

- KPI metrics computation (Churn %, ARPU, Revenue Risk)
- Statistical modeling & correlation analysis

- Seaborn visualization (Heatmaps, Pairplots, Catplots)

Data Schema & Ingestion

The analysis ingests three primary tables from `customer_churn.db`:

Table Name	Raw Columns Ingested	Primary/Foreign Keys
<code>db_customer</code>	<code>customerid</code> , <code>name</code> , <code>country</code> , <code>state</code> , <code>gender</code> , <code>dob</code> , <code>interests</code> , <code>pincode</code>	PK: <code>customerid</code>
<code>db_subscription</code>	<code>customerid</code> , <code>subscription_start_date</code> , <code>subscription_type</code> , <code>renewal_date</code> , <code>plan_type</code> , <code>contract_type</code> , <code>cancellation_date</code> , <code>cancellation_reason</code> , <code>monthly_charges</code> , <code>cltv</code> , <code>churn_score</code>	PK/FK: <code>customerid</code>
<code>db_support</code>	<code>customerid</code> , <code>complaint_date</code> , <code>escalations</code> , <code>csat_score</code> , <code>col_1</code> , <code>comment</code>	FK: <code>customerid</code>

Data Cleaning & Transformation Operations

1. `db_customer` Cleaning:

- **Renaming:** `name` renamed to `customer_name` for clarity.
- **Column Dropping:** Removed sparse/unnecessary attributes `interests` and `pincode`.
- **Data Type Conversion:** Converted `dob` to `datetime64[ns]`.
- **Standardization:** Standardized values in `gender` (`Men` -> `Male`, `Women` -> `Female`).
- **Missing Value Imputation:** Filled missing `country` values using state mappings (e.g., mapping states like `Delhi`, `Maharashtra` to `India`, and `Kathmandu` to `Nepal`).

2. `db_subscription` Cleaning:

- **Date Conversions:** Converted `subscription_start_date`, `renewal_date`, and `cancellation_date` into datetime structures.

3. `db_support` Cleaning & Deduplication:

- **Column Dropping:** Dropped empty column `col_1` and unstructured `comment` field.
- **Aggregation & Deduplication:** Calculated total complaints per customer (`complaint_count`). Sorted by `complaint_date` and dropped duplicates to avoid row multiplication during merging, maintaining a strictly 1:1 join relationship with the primary dataset.

Feature Engineering

- **`churn_flag`** : Binary flag indicating whether `cancellation_date` is populated.
- **`tenure_days`** : Calculated duration of customer tenure (difference between cancellation date/current date and subscription start date).
- **`churn_risk`** : Binned categorical risk status derived from `churn_score` :
 - `low` : `churn_score` < 50
 - `med` : `50 <= churn_score < 70`
 - `high` : `churn_score >= 70`

Core Findings & Business KPIs

Metric	Calculated Value	Business Interpretation
Total Customers	21	Total unique customer base evaluated.
Churn Rate	28.57%	Approximately 28.57% of customers canceled their subscriptions.
Retention Rate	71.43%	Portion of the customer base retained.
ARPU	\$18.85	Average Revenue Per User per month.
Average Tenure	1,517.29 Days	Average lifespan of active/churned accounts (~4.1 years).

Revenue at Risk	\$73.94	Direct monthly revenue lost due to churned accounts.
Escalation Rate	19.00%	Proportion of support complaints escalated to higher tiers.
Avg Complaints / User	0.43	Overall complaint rate across the total user base.

Segmented Insights

1. Churn by Subscription Plan Type

The data shows a strong inverse relationship between tier level and churn likelihood:

- **Basic Plan:** 60.00% churn rate (highest attrition rate across all tiers).
- **Standard Plan:** 22.22% churn rate.
- **Premium Plan:** 14.29% churn rate (strongest customer retention).

2. Plan Revenue & User Breakdown (Pivot Analysis)

Plan Type	Customer Count	Total Monthly Charges	Churn Rate (%)
Basic	5	\$52.95	60.00%
Standard	9	\$123.91	22.22%
Premium	7	\$218.93	14.29%

3. Geographic Observations

- **High Churn Regions:** **Karnataka** and **Meghalaya** exhibit elevated churn rates in the sample population compared to other regions like Delhi, Maharashtra, and Rajasthan.

Correlation & Exploratory Visualizations

- **Correlation Analysis:** Categorical encoding of ordinal parameters (`plan_type` , `contract_type` , `churn_risk` , `escalations`) shows a strong positive correlation

between support ticket **escalations**, high **churn_score** , and final **churn_flag** outcomes.

- **Time Series Trend:** Cancellations peaked during the mid-to-late months of 2024 (spiking around September 2024).
 - **Multivariate Categorical Plots (FacetGrid):** Indicates that lower tier plans combined with low customer satisfaction (CSAT) scores and escalated support tickets are primary drivers for high churn risk classification.
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Actionable Business Recommendations

1. **Revamp Basic Tier Offerings:** Because the **Basic Plan** exhibits a 60% churn rate, re-evaluate its feature set or onboarding experience to improve initial value delivery.
2. **Targeted Support Escalation Protocols:** Since support escalations correlate strongly with account cancellations, implement immediate outreach for any customer issuing an escalated ticket.
3. **Proactive Intervention via Risk Scoring:** Utilize the engineered **churn_risk** levels (**high** / **med**) to trigger automated retention discounts or success manager check-ins before final cancellation occurs.
4. **Focus Regional Success Initiatives:** Investigate localized service quality issues or regional competition in markets showing high churn rates (e.g., Karnataka).